

Effects of chitosan- and alginate-encapsulated olive leaf extract on performance, selected blood parameters, and antioxidant activity in broiler diets

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ABSTRACT

This study was conducted to determine the effects of supplementing broiler diets with olive leaf extract (OLE) and chitosan- and alginate-coated olive leaf extract (C-OLE) on performance; internal organ weights; selected blood serum biochemical values; breast- and thigh-meat pH; breast-meat MDA values; intestinal pH; tibia and femur morphometric characteristics (width, length, and weight); and dry matter levels and ammonia concentrations in litter and feces. In this study, OLE (1.5%) was subsequently encapsulated with alginate (1%) and chitosan (1%). After drying, the alginate- and chitosan-coated olive leaf extract was added to broiler rations at 1%. The research was conducted over 37 days with 105 one-day-old male broiler chicks. The chicks were randomly allocated to three groups: one control group and two treatment groups. The control group received a basal diet without supplementation. In contrast, one treatment group was fed a diet supplemented with OLE, and the other was fed a diet supplemented with C-OLE; each supplement contained 125 mg/kg oleuropein. On day 37, statistically significant differences were observed between the treatment groups and the control group in body weight, body weight gain, feed intake, feed conversion rate, the European Production Efficiency Factor (EPEF), and some organ weights ($P < 0.05$). In conclusion, encapsulated olive leaf extract containing 125 mg/kg oleuropein may improve the bioavailability of phenolic compounds and may represents a promising natural feed additive for broiler production.

Keywords: Alginate, Broiler, Chitosan, Encapsulation, Olive leaf extract.

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Introduction

Olive leaf extract (OLE) has recently been employed as a natural antioxidant because of its phenolic compounds. Olive leaves provide strong antioxidant activity due to phenolic compounds such as oleuropein, hydroxytyrosol, verbascoside, apigenin-7-glucoside, and luteolin-7-glucoside present in their structure (8, 41). Oleuropein, one of the main polyphenolic antioxidants found in olive leaves, is the most active phenolic compound in the leaf (36). It has been determined that this phenolic compound has natural antioxidant effects in both *in vitro* (11, 40) and

in vivo (30) studies. Oleuropein has also been reported to exhibit antimicrobial, antiviral, anti-inflammatory, and hepatoprotective effects.

In recent years, preference has been given to delivering active substances to the small intestine without exposing them to environmental factors or the acidic conditions of the stomach. For this purpose, studies have been conducted to control the release of phenolic compounds in OLE and to protect their antioxidant properties through encapsulation methods (21). For encapsulation, various polysaccharide-protein hydrogel

carriers are generally used. In the case of OLE, the alginate–chitosan encapsulation method has been reported to be a more effective option for protecting the stability of phenolic compounds, enhancing their bioavailability, and masking the strong odor and taste of aromatic extracts (20).

This study aimed to determine the effects of supplementing broiler diets with OLE and chitosan- and alginate-coated olive leaf extract (C-OLE) on performance, selected blood parameters, and antioxidant capacity. In addition, the use of C-OLE in broiler diets in Türkiye contributes to the novelty of this research. This study aims to generate data in this field and contribute to future commercial production processes.

Materials and Methods

Birds Husbandry and Diets: The study lasted 37 days and included 105 one-day-old male broiler chicks. The chicks were divided into three groups, each of which was further divided into five subgroups of seven chicks. The three groups consisted of one control group and two treatment groups. Ad libitum feeding and water were provided to both the control and treatment groups throughout the trial. The diets were prepared for all groups during the starter, grower, and finisher periods, as shown in Table 1. In the treatment groups, 125 mg/kg oleuropein was added to the

basal diet provided by 1.096 g/kg of OLE, which contained 114 g/kg oleuropein. One treatment group received 125 mg/kg oleuropein in its standard form and the other group received in oleuropein coated form.

Encapsulation of Olive Leaf Extract and Determination of Total Phenolic Substances: Initially, the best conditions for the encapsulation process were determined. Under these conditions, 1.5% OLE powder, 1% alginate, 1.7 M CaCl₂, and 1% chitosan solutions were used. Under the conditions of 700 Hz frequency, 200 bar pressure, and 700 V flow, olive leaf extract was encapsulated using an Encapsulator Device (BUCHI B-390). For stabilization, the alginate-coated microcapsules were kept at room temperature for 15 minutes. Then, these microcapsules were kept in 1% chitosan solution for 30 minutes to provide an additional coated layer. After the process, microcapsules were washed with deionized water and then dried.

The phenolic substance content of OLE and C-OLE was determined according to the spectrophotometric Folin–Ciocalteu method (18, 35). The absorbance was measured using a Shimadzu UV-1208 spectrophotometer at 765 nm. The calibration curve was obtained using gallic acid standards, and the results were expressed as gallic acid equivalents (GAE, mg/g).

Table 1. Ingredient composition, analyzed chemical composition (%), and metabolizable energy content (kcal/kg) of basal diets.

Ingredients	Starter 0-14 Days	Grower 15-28 Days	Finisher 29-37 Days
Maize	51.00	52.25	56.45
Full Fat Soybean	19.62	18.00	14.00
Vegatable Oil	1	2	3
Soybean Meal	24	24	23
Dicalcium phosphate	2.40	2.00	2.00
Limestone	0.8	0.85	0.85
Sodiumbicarbonate	0.1	0.1	0.1
Salt	0.25	0.25	0.25
DL-Methionine	0.37	0.25	0.15
L-lysine	0.2	0.10	0.0
Vitamin Premix*	0.10	0.10	0.10
Mineral Premix**	0.10	0.10	0.10
Anticoccidial	0.06	0.0	0.0
Total	100.00	100.00	100.00
Analyzed values			
Crude Protein	22.01	21.56	20.03
Metabolizable Energy	3099	3158	3219
Calcium	1.01	0.92	0.91
Phosphorus	0.5	0.44	0.44

*Vitamin premix in per kg of diets: Vitamin A: 11.000 IU, Vitamin D3: 3.500 IU, Vitamin E: 100 mg, Vitamin K3: 3 mg, Vitamin B1: 3 mg, B2: 6 mg, B6: 4 mg, B12: 0.02 mg, Niacin: 35 mg, Folic acid: 1.5 mg, Vitamin H: 0.2 mg, Vitamin B5: 15 mg. **Mineral premix in per kg of diets: Manganese: 120 mg, Zinc: 110 mg, Copper: 30 mg, Iron: 50 mg, Cobalt: 0.5 mg, Iodine: 2 mg, Selenium: 0.3 mg. *** Metabolizable energy content of diets was calculated according to the equation of Carpenter and Clegg (9).

Traits Measured: During the trial, body weight (BW) and feed intake (FI) were recorded for each chick on days 0, 7, 14, 21, 28, and 37. Feed conversion rate (FCR) was calculated as feed intake per kg of body weight gain (BWG). Carcass weight was measured after slaughter. Carcass yield (%) was determined at the end of the trial using the following formula. Carcass yield (%) = (body weight (g) on 37th day / carcass weight (g) on 37th day) X100. European Production Efficiency Factor (EPEF) was calculated per pen at 37 days using the following formula. EPEF, % = [(Body weight (kg)) * (Number of total alive chick / number of total chick at initial) * 100] / [(Trial period (day)) * FCR * 100]. At the end of the trial, two broilers per pen were randomly selected. After blood samples (a total of 30) were collected from the wing vein into vacuum tubes, the birds were slaughtered.

Measurement of Biochemical Parameters in Serum Samples: Levels of high-density lipoprotein (HDL), low-density lipoprotein (LDL), triglycerides (TG), total cholesterol (TC), alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase (ALP), and immunoglobulin G (IgG) were measured in serum samples collected at the end of the trial using an auto-analyzer and commercial analysis kits in a private laboratory.

Determination of Internal Organ Weights and Intestinal pH: The internal organs were removed from the carcass (two slaughtered animals from each subgroup), and the weights of the heart, liver, spleen, proventriculus, gizzard, abdominal fat, and bursa Fabricius were measured using a precision balance (± 10 mg sensitivity). The weights of the internal organs were divided by their live weights before slaughter to calculate relative organ weights.

Determination of pH Values of Breast and Thigh Meat and Measurement of Tibia and Femur Properties: After the left breast and thigh meat were separated from the carcass according to the TSE method (39), the samples were stored at +4°C for 24 hours. Then, the pH value was measured at five different points using a pre-calibrated meat probe pH meter (Mettler Toledo Seven Go Portable) (3). Following the pH measurement, the left thigh meat was separated from the femur and tibia bones. The femur and tibia bones, separated from the muscles, were weighed using a precision balance with a ± 10 mg sensitivity (Sartorius). Then, the width (the average of the widest and narrowest parts) and length (measured between the most distal points) were measured using a digital caliper (Mitutoyo 500-184-30 ABS), with measurements taken from the same reference points on each bone.

Determination of Breast Meat Malondialdehyde (MDA) Value: Two pieces of breast meat samples were taken at

slaughter from each animal. The TBARS method was used to determine the lipid oxidation level. The absorbance was measured at 530 nm using a spectrophotometer (Shimadzu UV-1208). To calculate TBARS values, absorbance values were multiplied by $K = 7.8$ and expressed as mg malondialdehyde/kg. TBARS values were calculated for samples stored at +4°C for one day and three days after slaughter (38, 34).

Determination of Dry Matter and Nitrogen Levels in Feces and Litter: At the end of the trial, feces were collected by placing plastic sheets beneath the sections, and litter samples were taken from three different parts of each section at approximately 3 cm below the litter surface. Dry matter was analyzed according to the AOAC method (4) in fecal and litter samples. The level of ammonia-nitrogen ($\text{NH}_3\text{-N}$) in feces and litter samples was determined by the indophenol blue method using a spectrophotometer (Shimadzu UV-1208) at a wavelength of 546 nm (10).

Statistical Analysis: All data were analyzed using one-way analysis of variance (ANOVA) to detect differences among group means. In addition, Tukey's multiple-range test was used to determine the significance of differences between groups. Chi-square analysis was used only to assess viability and mortality rates. Statistical analyses were performed using SPSS 22.0, and results were evaluated at the 95% confidence interval (37).

Results

Body weight gain, feed intake, and feed conversion rate are presented in Table 2. No significant differences were observed among the groups during the 0–21-day period ($P > 0.05$). However, during the 0–37-day period, the C-OLE group exhibited higher BWG and improved FCR compared with the control and OLE groups ($P < 0.05$). Feed intake was not significantly affected by dietary treatments throughout the experimental period ($P > 0.05$).

Slaughter weight and carcass weight differed significantly between the C-OLE and control groups ($P < 0.05$), whereas no significant differences in carcass yield were detected among the groups ($P > 0.05$) (Table 3). The EPEF was higher in the C-OLE group than in the OLE and control groups ($P < 0.05$) (Table 4). Although no statistically significant differences were observed in vitality rate among the groups, the highest value was recorded in the C-OLE group ($P > 0.05$, Table 4).

The average width, length, and weight of the tibia and femur were not significantly affected by the treatments ($P > 0.05$) (Table 5). Among the relative organ weights expressed per 100 g live weight, a significant difference was observed only for the proventriculus ($P < 0.05$). Except for the liver, lower relative organ weight values were generally observed in the C-OLE group than

in the control group. In addition, abdominal fat and gizzard weights expressed per 100 g live weight were lower in the OLE and C-OLE groups than in the control group. However, no significant differences were detected for the remaining organ parameters ($P>0.05$) (Table 6).

Serum biochemical parameters are presented in Table 7. No significant differences were observed among the groups in HDL, triglyceride, total cholesterol, ALT, AST, ALP, or IgG concentrations ($P>0.05$). In contrast, LDL concentration was significantly lower in the C-OLE group than in the control group ($P<0.05$).

The average pH values measured in different segments of the small intestine did not differ significantly

among the groups (Table 8). Similarly, no significant differences were observed in breast meat MDA levels or pH values (Table 9). Nevertheless, both MDA and pH values were numerically lower in the C-OLE group than in the other groups.

No significant differences were detected among the groups in the dry matter and nitrogen contents of feces and litter (Table 10). The concentrations of total phenolic compounds and oleuropein in OLE are presented in Table 11. The loading capacity, encapsulation efficiency, and *in vitro* release values of C-OLE determined in different parts of the digestive system are shown in Table 12.

Table 2. Body weight gain, feed intake and feed conversion rate of the groups.

Parameters	Control	OLE	C-OLE	SEM	P
0-7 days					
BWG (g)	110.51	107.57	109.25	1.21	0.64
FI (g)	141.52	149.11	139.97	2.46	0.28
FCR	1.28 ^b	1.39 ^a	1.28 ^b	0.01	0.02
8-14 days					
BWG (g)	238.28	249.43	252.50	2.75	0.07
FI (g)	391.86	388.10	388.69	6.37	0.97
FCR	1.65	1.56	1.55	0.01	0.40
15-21 days					
BWG (g)	438.622	438.628	447.11	4.47	0.70
FI (g)	602.70	646.77	609.26	11.57	0.26
FCR	1.37	1.48	1.36	0.02	0.11
22-28 days					
BWG (g)	612.93	660.13	655.84	11.70	0.19
FI (g)	956.95	992.97	983.38	12.99	0.53
FCR	1.78	1.51	1.50	0.07	0.26
29-37 days					
BWG (g)	955.58 ^b	908.89 ^b	988.60	13.11	0.03
FI (g)	1581.31	1525.66	1491.08	18.85	0.14
FCR	1.43 ^a	1.44 ^a	1.31 ^b	0.02	0.04
0-21 days					
BWG (g)	787.42	795.63	808.87	4.58	0.10
FI (g)	1136.08	1183.97	1137.92	15.06	0.36
FCR	1.44	1.49	1.41	0.02	0.15
0-37 days					
BWG (g)	2355.93 ^b	2364.66 ^b	2453.30 ^a	16.78	0.01
FI (g)	3674.34	3702.60	3612.38	26.58	0.39
FCR	1.56 ^a	1.57 ^a	1.47 ^b	0.06	0.02

OLE: Olive leaf extract, C-OLE: Chitosan- and alginate-coated olive leaf extract. BWG: Body weight gain, FI: Feed intake, FCR: Feed conversion rate. SEM: Standard error of mean. The difference between the means with different letters in the same row is significant ($P<0.05$).

Table 3. Slaughter weight, carcass weight and carcass yield values of the groups.

Parameters	Control	OLE	C-OLE	SEM	P
Slaughter weight, g	2359.77 ^b	2413.69 ^{ab}	2446.42 ^a	11.15	0.004
Carcass weight, g	1664.69 ^b	1711.47 ^{ab}	1722.08 ^a	10.27	0.047
Carcass yield, %	70.54	70.90	70.38	0.21	0.600

OLE: Olive leaf extract, C-OLE: Chitosan- and alginate-coated olive leaf extract. The difference between the means with different letters in the same row is significant ($P<0.05$). SEM: Standard error of mean.

Table 4. EPEF and vitality rate of the groups.

Parameters	Control	OLE	C-OLE	SEM	P
EPEF	414 ^b	411 ^b	458 ^a	8.01	0.01 (P)
Vitality rate, %	97.14	94.29	100.00	-	0.357 (χ^2)

OLE: Olive leaf extract, C-OLE: Chitosan- and alginate-coated olive leaf extract. EPEF: European Production Efficiency Factor, SEM: Standard error of mean. The difference between the means with different letters in the same row is significant (P<0.05).

Table 5. Mean values of tibia width, length and weight and femur width, length, and weight of the groups.

Parameters	Control	OLE	C-OLE	SEM	P
Tibia width mm	8.56	8.62	8.97	0.10	0.189
Tibia length, mm	91.90	92.02	93.10	0.41	0.446
Tibia weight, g	12.10	12.79	12.35	0.16	0.224
Femur width, mm	10.12	10.25	10.07	0.11	0.799
Femur length, mm	68.21	69.01	70.40	0.38	0.051
Femur weight, g	9.79	10.11	9.75	0.10	0.336

OLE: Olive leaf extract, C-OLE: Chitosan- and alginate-coated olive leaf extract. SEM: Standard error of mean.

Table 6. Relative organ weights (g/100 g live weight) of the groups.

Parameters	Control	OLE	C-OLE	SEM	P
Heart	0.49	0.50	0.48	0.01	0.747
Splen	0.12	0.12	0.09	0.01	0.103
Liver	1.99	2.01	2.04	0.03	0.795
Bursa Fabricius	0.19	0.19	0.17	0.01	0.332
Abdominal fat	1.17	1.13	0.95	0.05	0.207
Proventriculus	0.39 ^a	0.31 ^b	0.33 ^b	0.01	0.000
Gizzard	1.23	1.21	1.19	0.02	0.740

OLE: Olive leaf extract, C-OLE: Chitosan- and alginate-coated olive leaf extract. SEM: Standard error of mean. The difference between the means with different letters in the same row is significant (P<0.05).

Table 7. Levels of selected biochemical parameters in blood serum and plasma samples of the groups.

Parameters	Control	OLE	C-OLE	SEM	P
HDL, mg/dl	56.44	57.76	58.25	1.20	0.82
LDL, mg/dl	52.15 ^a	42.26 ^{ab}	36.11 ^b	2.44	0.01
TG, mg/dl	55.00	54.06	58.36	1.58	0.52
TC, mg/dl	115.47	105.81	107.29	2.04	0.11
ALT, IU/L	2.90	2.56	2.45	0.12	0.36
ALP, IU/L	2538.88	2821.51	2618.59	93.18	0.42
AST, IU/L	250.00	291.29	255.44	8.08	0.07
IgG, ml/dl	1229.49	1219.40	1110.00	53.08	0.61

OLE: Olive leaf extract, C-OLE: Chitosan- and alginate-coated olive leaf extract. SEM: Standard error of mean. HDL: High-density lipoprotein, LDL: Low-density lipoprotein, TG: Triglyceride, TC: Total cholesterol, ALT: Alanine aminotransferase, ALP: Alkaline phosphatase, AST: Aspartate aminotransferase, IgG: Immunoglobulin G. The difference between the means with different letters in the same row is significant (P<0.05).

Table 8. Mean intestinal pH values in duodenum, jejunum, and ileum.

pH value	Control	OLE	C-OLE	SEM	P
Duodenum	6.06	6.12	6.10	0.02	0.497
Jejunum	5.96	5.90	5.89	0.02	0.390
Ileum	6.19	6.26	5.93	0.07	0.115

OLE: Olive leaf extract, C-OLE: Chitosan- and alginate-coated olive leaf extract. SEM: Standard error of mean. The differences between the means of the groups are not statistically significant (P>0.05).

Table 9. MDA levels in breast meat and pH values in breast and thigh meat of the groups.

Parameters	Control	OLE	C-OLE	SEM	P
MDA, mg/kg	0.4347	0.5051	0.4017	0.11	0.110
Breast meat pH	5.88	5.87	5.83	0.01	0.234
Thigh meat pH	6.06	6.00	6.02	0.02	0.420

OLE: Olive leaf extract, C-OLE: Chitosan- and alginate-coated olive leaf extract. MDA: Malondialdehyde, SEM: Standard error of mean. MDA: Malondialdehyde.

Table 10. Dry matter and nitrogen levels in feces and litter of the groups.

Parameters	Control	OLE	C-OLE	SEM	P
Feces-DM, %	17.71	18.22	18.19	0.35	0.814
Feces-NH ₃ -N, mmol/l	3.68	3.70	3.78	0.23	0.986
Litter-DM, %	63.78	61.61	61.12	1.07	0.583
Litter-NH ₃ -N, mmol/l	12.55	14.13	13.19	0.48	0.443

OLE: Olive leaf extract, C-OLE: Chitosan- and alginate-coated olive leaf extract. SEM: Standard error of mean. DM: Dry matter, NH₃-N: ammonia-nitrogen.

Table 11. Total phenolic content and oleuropein concentration of OLE

Parameters	Total phenolic content, g/kg	Oleuropein, g/kg
OLE	227	114

OLE: Olive leaf extract.

Table 12. Preparation efficiency, loading capacity and encapsulation efficiency of C-OLE and invitro release percentages of C-OLE in different part of digestive system.

Parameters	Preparation efficiency	Encapsulation efficiency	Loading capacity	
C-OLE	100	90.06	53.28	
Parameters	Proventriculus	Gizzard	Duodenum	Small intestine
C-OLE	10.35	23.75	34.72	52.95
%	11.77	19.50	28.51	43.48

C-OLE: Chitosan- and alginate-coated olive leaf extract.

Discussion and Conclusion

The positive effects of olive-derived products on growth performance have been reported previously in poultry species. El-Sayed et al. (13) demonstrated improved body weight and body weight gain in broilers supplemented with olive oil, while Bahşi et al. (6) reported similar improvements in quails receiving oleuropein supplementation. Likewise, Natsir et al. (23) indicated that encapsulated phytochemical additives were more effective than their powder forms. The improved performance observed in the present study may therefore be associated with enhanced bioavailability of bioactive compounds through encapsulation.

Feed intake was not influenced by dietary treatments, which is in agreement with previous studies involving olive leaf extract supplementation in broilers and quails (6, 31, 33). Similarly, Shafey et al. (33) reported no effect of OLE on feed conversion rate, whereas Hafeez et al. (16) observed beneficial effects of encapsulated phytochemical

feed additives on feed efficiency. These findings suggest that encapsulation may improve the utilization of phytochemical compounds without affecting feed consumption.

The effect of olive-derived compounds on lipid metabolism has been documented previously. Andreadou et al. (2) reported reductions in plasma cholesterol and triglyceride concentrations following oleuropein supplementation in hypercholesterolemic rabbits. The lower LDL concentration observed in birds receiving C-OLE may be related to the improved bioavailability of phenolic compounds, as encapsulation has been shown to enhance the stability and absorption of aromatic and bioactive substances (22, 42).

Intestinal pH is an important determinant of digestive efficiency, microbial balance, and immune function in poultry. A slightly acidic intestinal environment favors the proliferation of beneficial microorganisms (29). Although no significant differences were observed among treatments, the intestinal pH values obtained in the present

study indicate that OLE supplementation did not adversely affect gut conditions.

Olive leaf polyphenols are known for their antioxidant, antimicrobial, and anti-inflammatory properties (24). However, their effects on humoral immune responses appear to be variable. In agreement with our findings, previous studies have reported limited effects on total IgG concentrations, although improvements in other immunoglobulin classes have been observed following olive leaf supplementation (28).

Lipid oxidation is a major factor influencing meat quality and shelf life. The absence of marked differences in MDA concentrations among treatments is consistent with previous reports (2, 15, 17). Nevertheless, the numerical reduction observed in birds receiving encapsulated olive leaf extract may suggest a tendency toward enhanced antioxidant protection, supporting the findings of Öztap and Küçükersan (26).

Meat pH is an important factor influencing the preservation of meat and meat products (1). Similarly, the lack of substantial effects on meat pH agrees with the observations of Paiva-Martins et al. (27), who reported that olive leaf supplementation did not alter meat pH values. In addition, no adverse effects were observed on litter quality parameters, indicating that OLE supplementation did not negatively affect excreta characteristics under the conditions of this study.

The total phenolic and oleuropein contents of the OLE used in the present study were comparable to values reported previously in Türkiye (5, 14, 25). Also our findings in oleuropein levels of OLE were in line with the studies that were conducted in Greece, France, and Spain (32, 19).

Furthermore, the encapsulation efficiency and release characteristics were consistent with findings reported by Chinoti et al. (12) and Bal (7). The *in vitro* release profile suggests that encapsulation may protect phenolic compounds from degradation in the upper digestive tract and promote their release in the intestine, thereby enhancing their biological effectiveness.

The findings of this study indicate that OLE containing 125 mg/kg oleuropein can be included in broiler diets without adverse effects. Encapsulation of olive leaf extract using alginate and chitosan appears to improve the utilization of bioactive compounds and may contribute to enhanced productive performance and favorable changes in selected blood parameters. The release characteristics of the encapsulated product further support its potential to increase the bioavailability of olive phenolics. Therefore, C-OLE may be considered a promising natural feed additive for broiler production. Further studies are needed to clarify its antioxidant and antimicrobial mechanisms and to evaluate its efficacy under different production conditions.

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Ethical Statement

This study was approved by the Ankara University Animal Experiments Local Ethics Committee (Approval No:2016-21-179).

Conflict of Interest

The authors declare that there is no conflict of interest.

Author Contributions

SK and NOH conceived and planned the experiments. NOH, SK and GÖ carried out the experiments. GÖ took the lead in writing the manuscript. All authors provided critical feedback and helped shape the research, analysis and manuscript.

Data Availability Statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declaration of Generative Artificial Intelligence (AI)

The authors declare that generative artificial intelligence (AI) and AI-assisted technologies were not used in the writing of this manuscript or in the preparation of figures.

Animal Welfare

The authors confirm that they have adhered to the ARRIVE Guidelines for the protection of animals used for scientific purposes.

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